

Artificial reef

This cagelike structure in the Indian Ocean, near Indonesia, is covered with calcium carbonate, which encourages threatened corals, such as elkhorn and staghorn, to grow, making the once-dense coral forests come alive again.

RESTORING REEFS

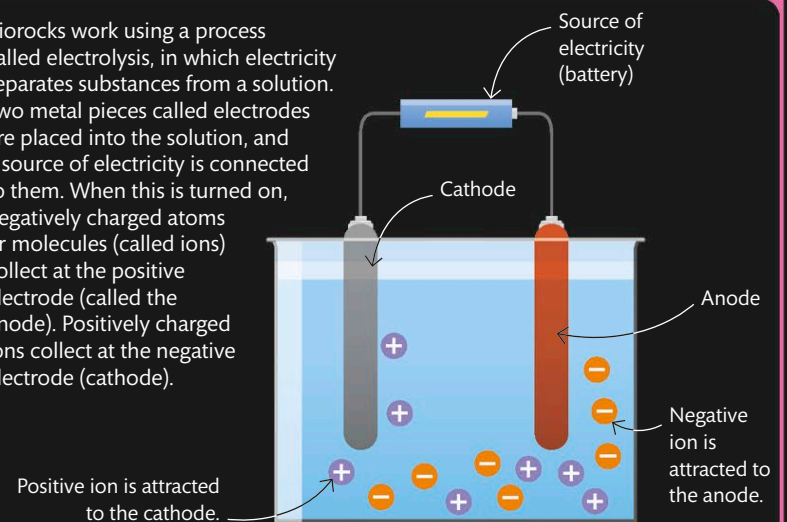
BIOROCKS

One of the most diverse ecosystems in the world, coral reefs are home to about one-quarter of the fishes in the ocean. However, in recent decades, water pollution, disease, and climate change have weakened and killed corals, threatening these important habitats.

This has inspired scientists to develop biorocks—a reef restoration method invented in Jamaica in the 1980s that uses a process called electrolysis to encourage new corals to grow.

ELECTROLYSIS

Biorocks work using a process called electrolysis, in which electricity separates substances from a solution. Two metal pieces called electrodes are placed into the solution, and a source of electricity is connected to them. When this is turned on, negatively charged atoms or molecules (called ions) collect at the positive electrode (called the anode). Positively charged ions collect at the negative electrode (cathode).



How biorocks work

In a biorock reef, fragments of living coral are attached to a metal cage. When the cage is connected to a source of electricity, it becomes a cathode, and electrolysis begins. The electrolysis causes chemical reactions in the water next to the cage. These allow calcium ions and carbonate ions from the seawater to join together, forming the mineral calcium carbonate on the cage. Corals then use this mineral to build their skeletons.

